

BHOOMISETU

“Connecting Land, Data and Decisions in Real Time”

TECHNICAL DOCUMENTATION

PROBLEM STATEMENT ID: 26016

Real-Time National Land Acquisition & Management System for End-to-End Digital Monitoring and Decision Support

THEME	Smart Automation
TEAM NAME	Hard Forkers
TEAM ID	144946
PROJECT NAME	BhoomiSetu

1

Team Details

1.1 Team Identity

Team Name	Hard Forkers
Problem Statement ID	26016 — Real-Time National Land Acquisition & Management System for End-to-End Digital Monitoring and Decision Support
Project Name	BhoomiSetu
Tagline	Connecting Land, Data and Decisions in Real Time

1.2 Team Members

Table 1. Team members

S. No.	Team Member Name	Role
1	Rishabh Mishra	Team Leader
2	Vaibhav Gupta	Team Member
3	Suraj Singh	Team Member
4	Rishabh Shisodiya	Team Member
5	Ashutosh Pandey	Team Member
6	Riya Aggarwal	Team Member

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2 Executive Summary

2.1 Background and Problem

Land acquisition underpins roads, railways, power, irrigation, industrial corridors and urban infrastructure. In India it is governed by the **RFCTLARR Act, 2013**, which links acquisition with Social Impact Assessment (SIA), fair compensation and Rehabilitation & Resettlement (R&R). Delivering these obligations requires a long chain of statutory steps, each owned by a different office.

In practice, the lifecycle still runs on state-specific procedures, paper files and disconnected portals. Central ministries lack a real-time view, district officers chase documents, project implementing agencies (PIAs) wait without visibility, and affected families cannot easily find out where their case stands.

VISION

A single, transparent, GIS-enabled national view of every land acquisition case, from proposal to R&R closure.

MISSION

Digitise the end-to-end lifecycle through a role-based, workflow-driven platform that connects land, data and decisions in real time, with AI-assisted decision support and proactive timeline monitoring.

2.2 Proposed Solution

BhoomiSetu is a unified digital platform for managing the complete land acquisition lifecycle. It combines digital case intake with GIS capture, a configurable **12-stage workflow engine**, AI/ML-assisted compensation anomaly detection, compensation and possession management, R&R tracking, multi-level monitoring dashboards and a citizen transparency view, all governed by role-based access control (RBAC) and tamper-evident audit trails.

2.3 Why Multi-Agency Coordination Is Essential

No single department can complete an acquisition. Each statutory stage depends on records, approvals or funds held by another body, so a shared case record and shared status are the foundation of the system.

Table 2. Agencies involved in a land acquisition case and their contribution

Agency / body	Contribution	Agency / body	Contribution
Central ministry / nodal agency	Policy oversight, national monitoring	Project Implementing Agency (PIA)	Project need, land requirement, funds
State revenue / land department	Appropriate-government approvals, state rules	Land records & survey departments	Cadastral maps, ownership records
District Collector / LAO	Notifications, hearings, award, possession	SIA unit & expert group	Social impact study and appraisal
Finance / treasury / payment systems	Fund release and compensation payment	R&R administration / authority	Entitlements, R&R delivery and grievance

2.4 Major Stakeholders and Key Capabilities

Stakeholders (six roles)	Key capabilities
- Ministry Analyst - State Government Officer - District Collector - Land Acquisition Officer (LAO) - PIA Officer - Citizen / Land Owner	- Digital case intake with document and compliance checks - GIS geo-tagging of parcels and project right-of-way 12-stage workflow with role-based routing and deadlines - AI/ML anomaly flags with reason codes for officer review - Compensation, geo-tagged possession and R&R tracking - National-to-citizen dashboards, alerts and MIS reports

2.5 Expected Transformation

Table 3. Expected transformation from current practice to BhoomiSetu (design intent)

From (current practice)	To (BhoomiSetu)
Paper files moving between offices	One digital case record with versioned, hash-protected documents
Manual hand-offs and follow-ups	Automated role-based routing with stage deadlines
Stage-wise opacity for officers and citizens	Real-time status dashboards at every administrative level
Boundary ambiguity and disputes	Geo-tagged parcels validated against cadastral and project layers
Reactive tracking of statutory time limits	Proactive alerts for pending, delayed and expiring milestones
Unexamined valuation inconsistencies	Explainable anomaly flags routed to officer review

3 Problem Statement & Existing System

Problem Statement 26016 asks for a real-time national system for end-to-end digital monitoring and decision support in land acquisition. The existing arrangement is characterised by the following challenges.

Table 4. Challenges in the existing land acquisition system

#	Challenge	Observed effect
1	Fragmented systems	Separate portals, registers and spreadsheets; no shared case record
2	Manual documentation	Proposals, notices, survey sheets and awards prepared on paper; errors and loss
3	Physical file movement	Approvals stall with the file; status is invisible to others
4	State-specific workflows	Formats and procedures differ, preventing a comparable national picture
5	Data duplication	Parcel, owner and project data re-entered at every stage and office
6	Delayed approvals	No automated routing or escalation; pending items go unnoticed
7	Limited transparency	Landowners and oversight bodies lack reliable case-level status
8	Poor inter-agency coordination	Ministry, state, district and PIA work without a shared view
9	Compensation tracking issues	Assessed vs. disbursed amounts hard to reconcile; valuation outliers undetected
10	Possession monitoring issues	Handover recorded on paper without geo-referenced evidence
11	R&R tracking issues	Entitlements, families and R&R milestones tracked inconsistently
12	Lack of real-time information	MIS data is periodic and compiled manually
13	GIS / boundary ambiguity	Survey, cadastral and project alignment data not reconciled; disputes follow
14	No proactive timeline monitoring	Statutory time limits tracked manually; breaches noticed after the fact

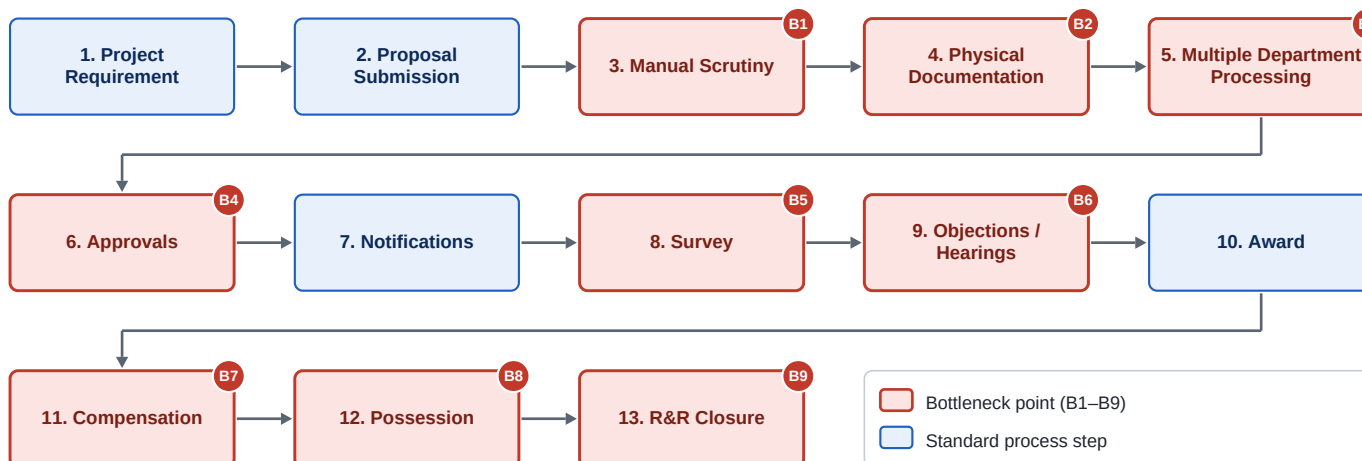


Figure 1. Existing Land Acquisition Process

Figure 1 shows the conventional sequential, paper-driven lifecycle from project requirement to R&R closure. Red boxes B1 to B9 mark the bottlenecks: manual scrutiny, physical file movement, multi-department processing, approval delays, survey ambiguity, objection handling, compensation tracking, possession evidence and R&R closure. Each hand-off depends on physical files with no common real-time status.

Table 5. Bottleneck key for Figure 1

B1	Manual scrutiny of paper proposals	B4	Approvals without routing or escalation	B7	Compensation and valuation tracking
B2	Physical documentation and file movement	B5	Survey and boundary ambiguity	B8	Possession without geo-evidence
B3	Multi-department processing, no shared status	B6	Objection and hearing record scatter	B9	Weak R&R closure tracking

4 BhoomiSetu Solution Overview

BhoomiSetu replaces fragmented, paper-driven coordination with one digital case record, one workflow and one national view. Seven solution pillars address the challenges identified in Section 3.

Table 6. BhoomiSetu solution pillars and the challenges they address

Pillar	Key functions	Challenges addressed
1. Case Intake & GIS Capture	Digital proposal intake, document upload with compliance checks, parcel capture and geo-tagging	Fragmentation, manual documentation, duplication, boundary ambiguity
2. 12-Stage Workflow Orchestration	Configurable stage engine with role-based routing, statuses and statutory deadlines	File movement, delayed approvals, state-specific workflows
3. AI/ML Compensation Anomaly Detection	Isolation Forest scoring, reason codes and officer review queue (decision support only)	Undetected valuation inconsistencies
4. Compensation & Possession Management	Award records, payment-status tracking, geo-tagged possession records	Compensation and possession tracking issues
5. Rehabilitation & Resettlement Tracking	Affected-family register, entitlement and milestone tracking until closure	R&R tracking issues
6. Monitoring & Governance	National to project dashboards, alerts, MIS reports, audit trails	No real-time information, weak coordination, no proactive monitoring
7. Citizen Transparency	Citizen view of own parcel, notices, award and R&R status	Limited transparency

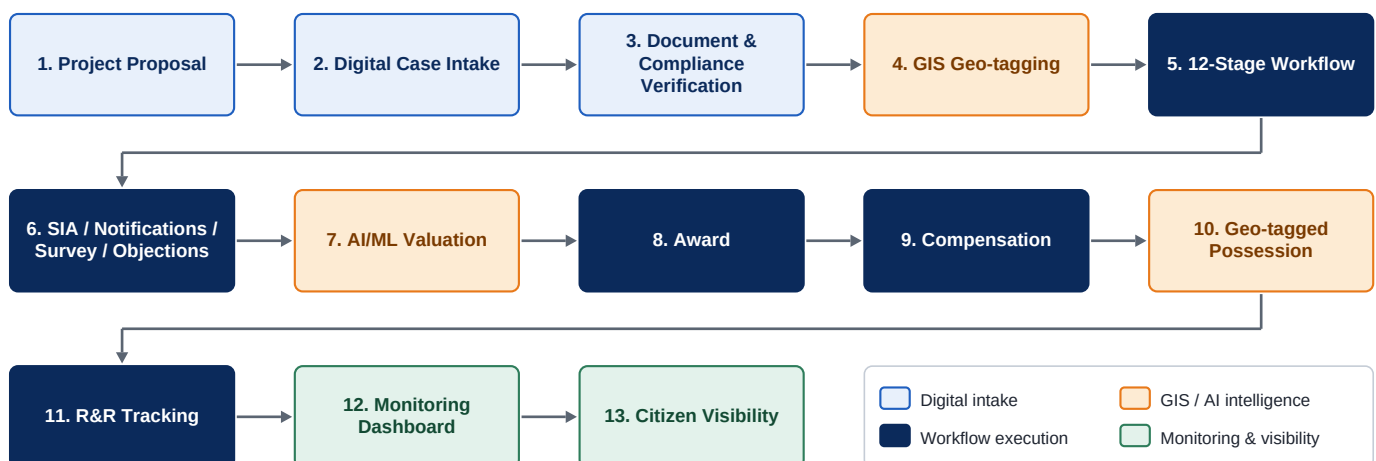


Figure 2. BhoomiSetu End-to-End Solution Flow

Figure 2 traces a land acquisition case through BhoomiSetu from proposal to citizen visibility. Blue steps are digital intake, orange steps apply GIS or AI/ML intelligence, navy steps execute the workflow and green steps provide monitoring and visibility. Every step writes to the same case record, so each stakeholder sees consistent, role-appropriate information.

4.1 Prototype Scope vs. Proposed Production Architecture

Aspect	Treatment in this document
Core platform	Web application, workflow engine, RBAC, GIS parcel views, anomaly detection module and dashboards are described as the BhoomiSetu prototype scope.
External government systems	State land records, cadastral maps, Bhuvan, PFMS, DILRMP, DigiLocker / eSign and LGD are marked Proposed / Integration Ready . No live integration is claimed.
Production deployment	Government cloud hosting, GeoServer-based map serving and scale-out are part of the proposed production architecture.

5 Stakeholders & Role-Based System

BhoomiSetu is organised around six system roles that mirror the institutional hierarchy of land acquisition. Each role has a defined responsibility and a restricted set of screens, records and actions.

Table 7. System roles, responsibilities and major system access

Role	Responsibility	Major system access
1. Ministry Analyst	Monitors national progress, compares states and projects, identifies stalled cases and reviews policy-level indicators.	National dashboard, state and project comparisons, MIS reports, read-only case summaries and audit views (national scope).
2. State Government Officer	Scrutinises proposals, evaluates SIA outcomes and grants state-level approvals; oversees all districts in the state.	State dashboard, proposal scrutiny and approval queues, SIA evaluation records, district-wise progress and delay reports.
3. District Collector	Authorises notifications, declarations and awards; presides over objections and hearings; oversees possession and R&R in the district.	District dashboard, authorisation queues, hearing schedules, award approval, possession and R&R oversight, district alerts.
4. Land Acquisition Officer (LAO)	Runs day-to-day case execution: intake, survey, notices, valuation review, award preparation, compensation and possession records.	Full operational access to assigned cases: parcel and document management, survey entry, valuation and anomaly review queue, compensation and possession entry.
5. PIA Officer	Submits the land requirement and supporting documents, provides funds information and tracks project readiness and land availability.	Own-project proposal forms, document uploads, project dashboard, proposed vs. acquired land and possession status for own projects.
6. Citizen / Land Owner	Views own parcel details and case status, files objections and tracks notices, award, compensation and R&R entitlement.	Citizen dashboard (own records only), public notices, objection submission, award and R&R status, grievance tracking.

5.1 How RBAC Delivers the Right View to Each Stakeholder

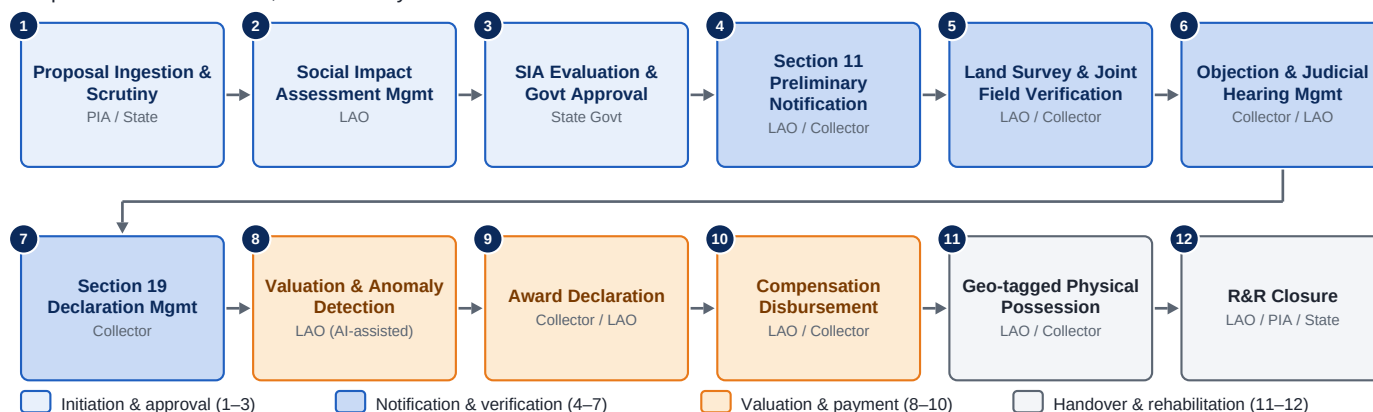
- **Role-bound permissions:** every API request is authorised against the caller's role, so a stakeholder can only open screens and perform actions defined for that role.
- **Jurisdiction scoping:** data is filtered by national, state, district or project scope; a District Collector sees only the assigned district, a PIA Officer only own projects.
- **Own-record access for citizens:** a citizen can see only the parcels and cases linked to their own verified identity, protecting privacy.
- **Separation of duties:** preparation (LAO), authorisation (Collector / State) and oversight (Ministry) are performed by different roles, and each action is audit-logged.

Table 8. Role scope of data visibility

Ministry Analyst	State Govt Officer	District Collector	LAO	PIA Officer	Citizen / Land Owner
National	State	District	Assigned cases	Own projects	Own parcels / cases

6 12-Stage Land Acquisition Workflow

The workflow engine moves each case through twelve ordered stages. A stage opens only when the previous stage output is recorded, the responsible role is notified, and statutory deadlines are tracked on the case timeline.



Small text under each stage name = primary responsible role.

Figure 3. 12-Stage BhoomiSetu Workflow

Figure 3 presents the twelve stages grouped into four phases, with the primary responsible role shown under each stage name. Each stage consumes the output of the previous one and produces a recorded outcome that unlocks the next stage. Stage 8 uses AI/ML only to support the LAO's review.

Table 9. Stage-wise purpose, input, process, responsible role, output and next stage

Stage	Purpose	Input	Process	Responsible role	Output → Next
1 Proposal Ingestion & Scrutiny	Admit a complete, valid proposal	Project proposal, land requirement, parcel list, documents	Digital intake, document and compliance checks, GIS capture, scrutiny	PIA Officer (submit); State Officer (scrutiny)	Scrutinised, geo-tagged case → 2
2 SIA Management	Capture the social impact study	Scrutinised case, SIA scope	Track SIA commissioning, study, public hearing records, report upload	LAO; State Officer	SIA report on record → 3
3 SIA Evaluation & Govt Approval	Decide on appraisal and approval	SIA report, appraisal remarks	Review, record observations, approve or return with remarks	State Govt Officer	Approval order → 4
4 Section 11 Preliminary Notification	Publish intent to acquire	Approved case, parcel schedule	Generate and publish notification, notify owners, start statutory clock	LAO; Collector authorises	Notification record and timeline → 5
5 Land Survey & Joint Field Verification	Verify parcels, owners and assets	Notified parcels, cadastral and GIS layers	Joint field survey, record area, owner and structures, reconcile boundaries	LAO; Collector oversees	Verified parcel register → 6
6 Objection & Judicial Hearing Mgmt	Handle claims and objections	Objections filed by owners and interested persons	Register objections, schedule hearings, record decisions	Collector / LAO; citizen files	Disposed objections → 7
7 Section 19 Declaration Mgmt	Make the final declaration	Disposed objections, verified parcels	Prepare, authorise and publish declaration; check against Section 11 timeline	Collector; LAO	Declaration record → 8
8 Valuation & Anomaly Detection	Assess compensation and flag outliers	Parcel, land-use, transaction and circle-rate data	Valuation, Isolation Forest scoring, reason codes, officer review	LAO (AI is decision support)	Reviewed valuation → 9
9 Award Declaration	Declare the award	Reviewed valuation, ownership details	Compile award statement, authorise, publish, notify beneficiaries	Collector; LAO	Award record → 10
10 Compensation Disbursement	Pay entitled persons	Award, beneficiary details	Track payment status and reconcile; PFMS linkage is proposed	LAO; Collector; PIA (funds)	Disbursement ledger → 11
11 Geo-tagged Physical Possession	Take possession with evidence	Disbursement confirmation	Record possession with geo-tag, date and evidence; update acquired land	LAO; Collector	Possession record → 12
12 R&R Closure	Deliver R&R entitlements	Affected-family register, R&R scheme	Track entitlements per family, verify milestones, close case	LAO; PIA; State Officer	Closed case (end of lifecycle)

7

GIS & Geo-Tagging

Land acquisition is inherently spatial. BhoomiSetu treats every parcel as a geo-referenced record so that proposed land, notified land, acquired land and possession status can be seen on a map and validated spatially.

Table 10. GIS capabilities of BhoomiSetu

Capability	Description
Land parcel mapping	Parcel records linked to survey number, owner, area and project, displayed as map polygons.
Geo-tagging	Coordinates and boundaries attached to parcels, survey points and possession evidence.
Cadastral boundaries	Cadastral layer overlay from state / national sources (proposed integration).
Project Right-of-Way	Project alignment or area layer for overlay against affected parcels.
Satellite imagery and base maps	Satellite and base-map backdrops for context and field verification (Bhuvan integration proposed).
Parcel visualization	Colour-coded parcel status views at project, district and state level.
Proposed vs. acquired land	Comparison of project land requirement with parcels notified, awarded and acquired.
Possession status	Geo-tagged possession records with date and evidence.
Spatial validation	Coordinate checks and overlap / containment checks using spatial queries.
Boundary verification	Reconciliation of surveyed, cadastral and project boundaries to reduce ambiguity.

Table 11. GIS technologies and their role

Technology	Role in BhoomiSetu	Status
Leaflet	Interactive web maps (React-Leaflet) for parcel and dashboard views	Prototype stack
PostGIS	Spatial storage and queries on PostgreSQL (geometry, overlay, validation)	Prototype stack
GeoServer	Standards-based publishing of spatial layers (e.g., WMS / WFS)	Proposed production
QGIS	Desktop GIS for preparing and validating spatial data by GIS teams	Supporting tool
BhuNaksha	Cadastral map source for state parcel boundaries	Proposed / Integration Ready
Bhuvan (ISRO)	Satellite imagery and national geo-portal layers	Proposed / Integration Ready

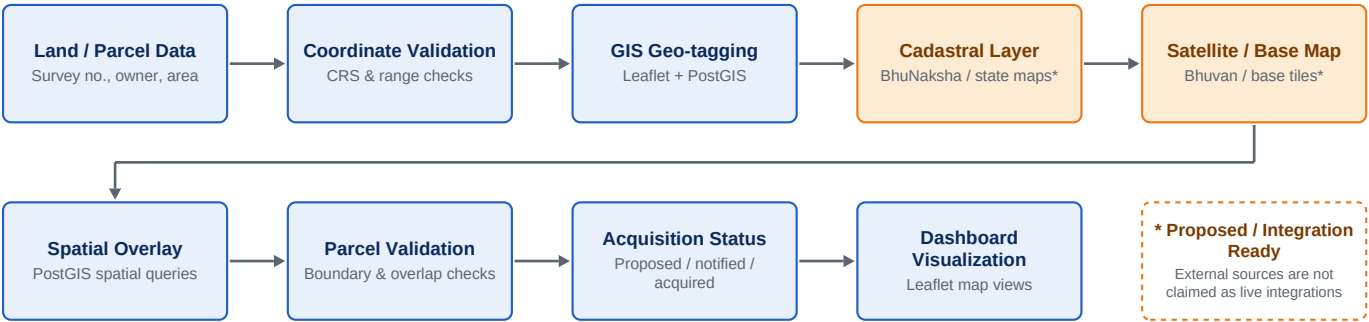


Figure 4. GIS Parcel Processing Flow

Figure 4 shows how a parcel record is processed from raw land data to dashboard visualisation. Coordinates are validated and geo-tagged, overlaid on cadastral and base-map layers, validated spatially and finally assigned an acquisition status shown on the map. Orange boxes marked with an asterisk depend on external sources that are proposed, not live.

8 AI/ML Compensation Anomaly Detection

The AI/ML module screens valuations for unusual patterns so that officers can concentrate on cases that deserve closer review. It is implemented in Python (Pandas, NumPy, scikit-learn) and uses the **Isolation Forest** algorithm.

Table 12. Components of the anomaly detection module

Component	Description
Land transaction data	Historical sale / registration data, parcel attributes, land use, location and area used as comparison context.
Circle rate and market rate baseline	Government circle rate and observed market rate form the reference against which proposed compensation is compared.
Data preprocessing and feature engineering	Cleaning, missing-value handling, scaling and derived features such as value per unit area, ratio to circle rate and deviation from locality baseline.
Compensation valuation	Compensation computed from the baseline and applicable rules; the valuation is the input to anomaly screening.
Isolation Forest	Unsupervised ensemble method that isolates observations by random partitioning; outliers need fewer splits and receive higher anomaly scores.
Anomaly detection	Scores compared with a configurable review threshold; cases above it are flagged.
Explainability and reason codes	Each flag carries the features and comparisons that contributed to it, expressed as officer-readable reason codes.
Officer review	Flagged cases enter the LAO review queue; the officer confirms, adjusts or documents a justification. Unflagged cases are auto-cleared.

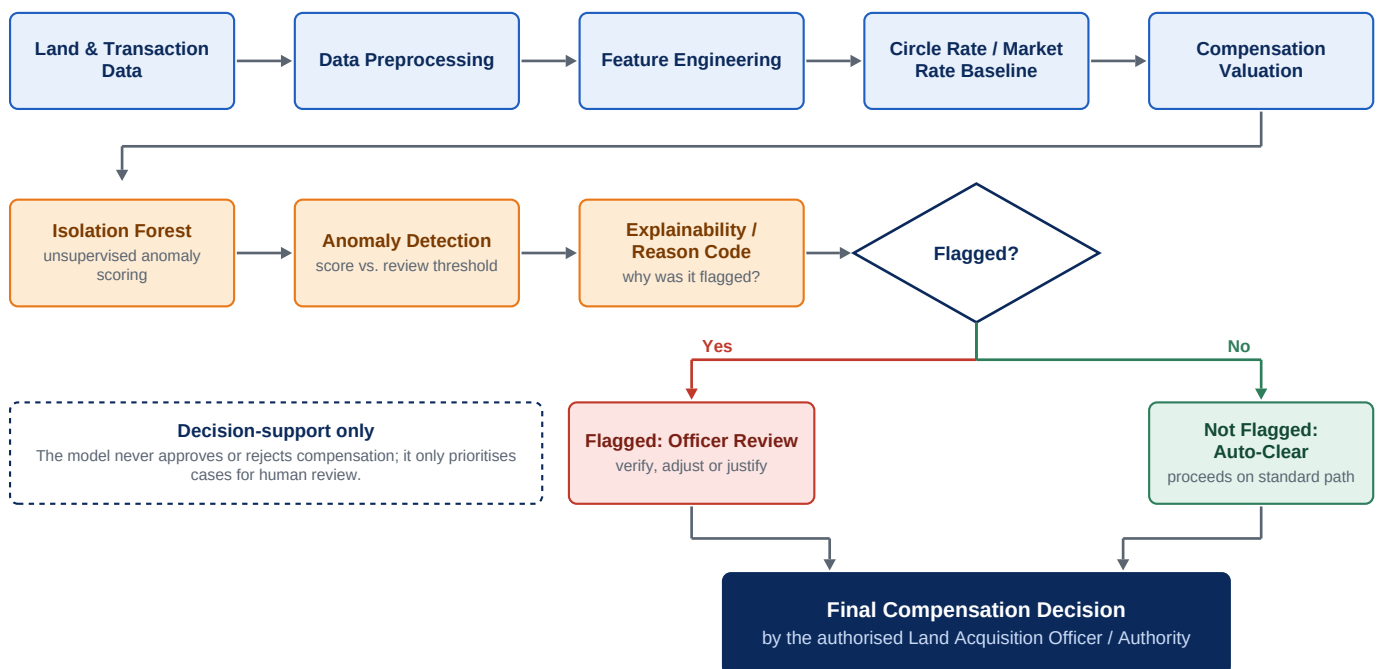


Figure 5. AI/ML Anomaly Detection Pipeline

Figure 5 shows the pipeline from land and transaction data to the final compensation decision. Valuation outputs are scored by Isolation Forest, explained through reason codes and routed either to officer review (flagged) or standard processing (not flagged). In both branches the final decision rests with the authorised officer.

IMPORTANT: AI/ML is used as a decision-support mechanism. It does not independently approve compensation. Final approval remains with the authorized government authority / Land Acquisition Officer.

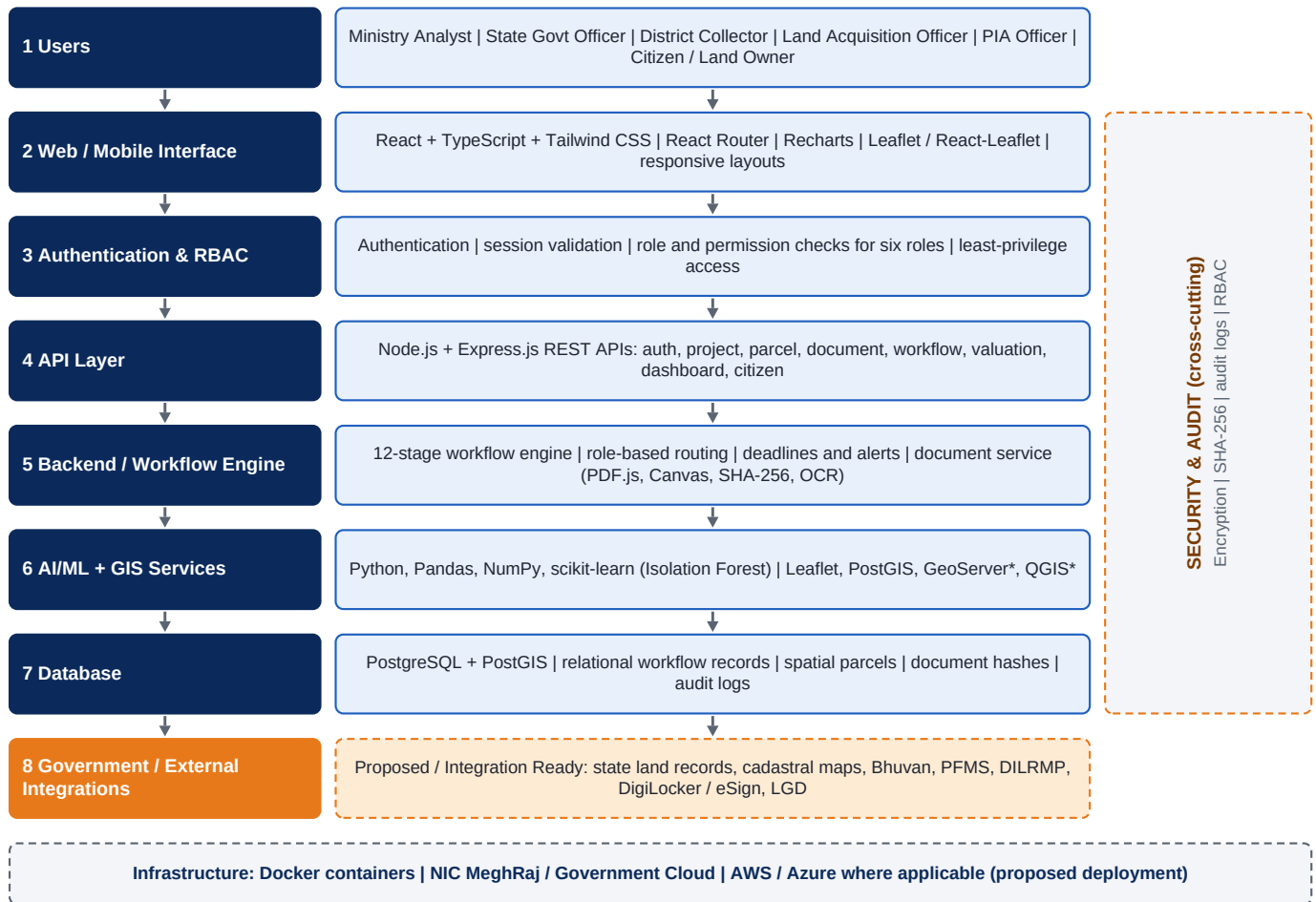
Table 13. Illustrative reason codes (examples of the explanation format)

Code	Illustrative meaning
RC-01	Proposed rate deviates strongly from the circle-rate baseline for the locality
RC-02	Proposed rate differs markedly from recent comparable transactions
RC-03	Unusual value-to-area or land-use combination compared with similar parcels

Model thresholds, features and reason codes are configurable and must be calibrated and validated on authorised government data before any operational use. No accuracy figures are claimed in this document.

9 Technical Architecture

BhoomiSetu follows a layered architecture in which every request passes through authentication and RBAC before reaching the API layer, workflow engine and data services. Security and audit logging cut across the layers.



* GeoServer and QGIS: GIS serving / desktop validation tools in the production architecture; external integrations are proposed, not live.

Figure 6. BhoomiSetu Layered Technical Architecture

Figure 6 shows the layers from users down to government and external integrations. Authentication and RBAC sit between the interface and the API layer, the workflow engine orchestrates AI/ML and GIS services, and PostgreSQL/PostGIS is the system of record. The integration layer is dashed because those integrations are proposed.

Table 14. Technology stack by layer

Layer	Technologies	Notes
Frontend	React, TypeScript, Tailwind CSS, React Router, Recharts, Leaflet / React-Leaflet	Role-based screens, charts, maps
Backend	Node.js, Express.js, REST APIs	Workflow engine and service endpoints
Database	PostgreSQL, PostGIS	Relational plus spatial data
AI/ML	Python, Pandas, NumPy, scikit-learn, Isolation Forest	Decision-support scoring only
GIS	Leaflet, GeoServer, QGIS	GeoServer / QGIS in production workflow
Document processing	PDF.js, HTML5 Canvas, OCR / document processing	Viewing, annotation, extraction
Security	RBAC, SHA-256, audit logs, authentication, encryption	Detailed in Section 11
Infrastructure	Docker, NIC MeghRaj / Government Cloud, AWS / Azure where applicable	Proposed deployment targets

10 Database & API Architecture

The conceptual data model keeps one authoritative record per project and parcel and attaches every statutory step, document and financial outcome to it. This is a conceptual structure; table and field definitions will be finalised during detailed design.

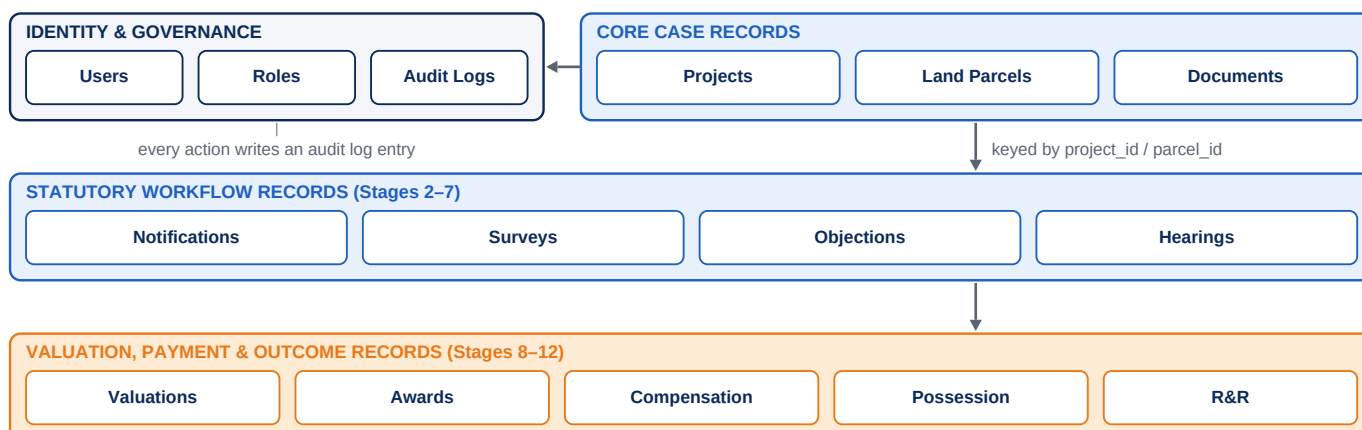


Figure 7. Conceptual Entity Structure

Figure 7 groups the fifteen major entities by purpose. Projects, land parcels and documents form the core case record; statutory records and outcome records hang from them by project and parcel identifiers, while audit logs record every action by users in their roles.

Table 15. Major conceptual entities

Entity	Purpose
Users / Roles	Identity, role assignment and jurisdiction scope
Projects	Acquisition projects, PIA, requirement, status
Land Parcels	Survey no., owner, area, geometry, status
Documents	Files, versions, SHA-256 hashes, metadata
Notifications / Surveys	Statutory notices and field survey records
Objections / Hearings	Claims, schedules, decisions
Valuations / Awards	Valuation inputs, flags, award details
Compensation / Possession / R&R	Payments, geo-tagged possession, entitlements
Audit Logs	Who did what, when, from where

Table 16. API categories (illustrative endpoint groups)

API category	Representative functions
Authentication APIs	Login, session validation, role resolution
Project APIs	Create / update projects, proposal submission, status
Land Parcel APIs	Parcel CRUD, geo-tagging, spatial queries
Document APIs	Upload, versioning, hash verification, retrieval
Workflow APIs	Stage transitions, assignments, deadlines, notifications
Valuation APIs	Valuation requests, anomaly scores, reason codes, review
Dashboard APIs	Aggregated indicators by national, state, district, project
Citizen APIs	Own-parcel view, notices, objections, status

Table 17. Interoperability (Proposed / Integration Ready)

System	Intended exchange	Status
State land records	Ownership and mutation data for parcels	Proposed / Integration Ready
Cadastral maps (BhuNaksha)	Parcel boundaries for GIS overlay	Proposed / Integration Ready
PFMS	Payment status for compensation tracking	Proposed / Integration Ready
Bhuvan	Satellite imagery and base layers	Proposed / Integration Ready
DILRMP	Digitised land record alignment	Proposed / Integration Ready
DigiLocker / eSign	Document retrieval and digital signing	Proposed / Integration Ready
LGD	Standard administrative-unit codes	Proposed / Integration Ready

11 Security, RBAC & Data Governance

Land records contain personal and financial information. BhoomiSetu applies least-privilege access, protects documents with cryptographic hashes and records every significant action so that the case history can be audited.

Table 18. Security and data governance controls

Control	How it is addressed
Role-Based Access Control	Permissions are attached to six roles and enforced on every API request.
Authentication	Users are identified before access; sessions are validated on each request.
Authorization and least privilege	Each role gets only the actions and jurisdiction scope it needs.
Citizen privacy	Citizens see only their own records; personal data is minimised in dashboards and reports, in line with the DPDP Act, 2023 principles.
Government data protection	Sensitive data is kept in controlled government / approved cloud infrastructure (proposed deployment).
Encryption	Encryption in transit and for sensitive data at rest is part of the production architecture.
SHA-256 document hashing	A hash is stored at upload; a changed file no longer matches its recorded hash.
Audit logs	Actions recorded with user identity, role, timestamp and IP address.
Version control	Documents and critical records keep versions rather than silent overwrites.
Tamper-evident records	Hashes and append-only audit entries make unauthorised changes detectable.
Secure document repository	Documents are stored centrally with access checks, not shared by e-mail or physical files.

Table 19. RBAC permission matrix (indicative defaults, configurable per state rules)

Function	Ministry Analyst	State Officer	District Collector	LAO	PIA Officer	Citizen
Proposal intake & scrutiny	V	A	V	F	O	-
SIA management & approval	V	A	V	F	V	P
Section 11 / 19 notifications	V	A	A	F	V	P
Survey & joint verification	V	V	A	F	V	V
Objections & hearings	V	V	A	F	-	O
Valuation & anomaly review	V	V	A	F	V	-
Award declaration	V	V	A	F	V	V
Compensation disbursement	V	V	A	F	V	V
Geo-tagged possession	V	V	A	F	V	V
R&R tracking & closure	V	V	A	F	V	V
Dashboards & MIS (scope)	National	State	District	Cases	Project	Own
Audit log access	V	V	V	-	-	-

Legend: A = approve / authorise; F = full operational create and update (assigned cases); O = create or update own records only (own project or own objection); V = view within jurisdiction or own records; P = public notices only; - = no access.

12 Dashboards, Monitoring & Alerts

Dashboards are computed from the same case records that officers update, so monitoring needs no separate reporting step. Indicators roll up from project to district, state and national level.

Table 20. Indicators tracked by BhoomiSetu

#	Indicator	What is tracked
1	Land proposed vs. acquired	Area required by projects against area notified, awarded and in possession
2	Notifications issued	Section 11 and Section 19 notifications by project, district and state
3	Awards declared	Number and status of awards and total award amount
4	Compensation assessed & disbursed	Assessed amount, disbursed amount, pending payments, anomaly flags
5	Possession status	Parcels and area handed over, geo-tag evidence, pending possession
6	R&R progress	Entitlements delivered, milestones met, pending R&R items
7	Affected & displaced families	Families identified, verified, compensated and resettled
8	Project-wise / state-wise progress	Stage distribution and progress comparison across projects and states
9	Timeline & milestone tracking	Statutory milestone dates, elapsed time and days remaining

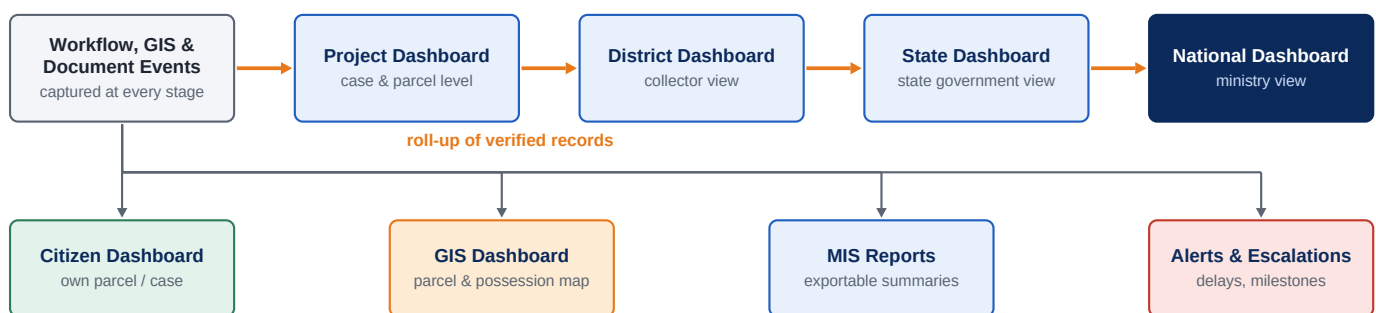


Figure 8. Monitoring Roll-up and Views

Figure 8 shows how workflow, GIS and document events captured at every stage roll up through project, district, state and national dashboards. The same data feeds the citizen dashboard, the GIS dashboard, MIS reports and the alert engine.

Table 21. Dashboard views

Dashboard	Audience	Key views
National	Ministry Analyst	State-wise comparison, stalled cases, national totals
State	State Government Officer	District-wise progress, approvals pending, delay report
District	District Collector, LAO	Case pipeline by stage, hearings, awards, possession, alerts
Project	PIA Officer, LAO	Proposed vs. acquired land, timeline, milestone status
Citizen	Citizen / Land Owner	Own parcel status, notices, award, compensation, R&R
GIS	All officer roles (scoped)	Parcel status map, possession map, project overlay
MIS reports	Ministry, State, District	Exportable periodic and ad-hoc summaries

Table 22. Alerts and escalations

Alert	Trigger	Notified role
Pending approval	A case waits in an approval queue beyond its configured period	Approver; next-level supervisor
Delayed case	A stage exceeds its expected duration	LAO; Collector
Statutory milestone	A statutory time limit is approaching or has lapsed	Collector; State Officer
Anomaly review pending	A flagged valuation has not been reviewed	LAO; Collector

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Feasibility & Implementation

Table 23. Feasibility assessment

Dimension	Assessment
Technical feasibility	Built on mature technologies: React / Node.js web stack, PostgreSQL / PostGIS, established GIS tooling (Leaflet, GeoServer, QGIS), REST APIs and container-based cloud deployment.
Economic feasibility	Open-source components limit licensing cost. Digital workflows and centralised data are expected to reduce paperwork, repeated data entry and administrative overhead. No cost figures are claimed here.
Operational feasibility	Role-based workflows, automated routing and real-time monitoring map onto existing offices and roles, so users keep familiar responsibilities.
Security feasibility	RBAC, encryption, audit trails and secure authentication are standard, well-understood controls suitable for government deployment.
Scalability	Designed for state-level deployment first, then multi-state and national deployment, with multilingual support planned. The state-specific rule configuration avoids code forks per state.

13.1 Implementation Roadmap

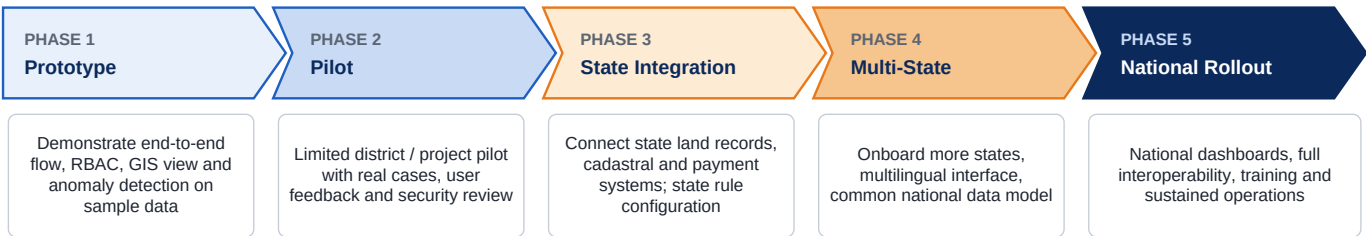


Figure 9. BhoomiSetu Implementation Roadmap

Figure 9 shows the planned five-phase progression from prototype to national rollout. Each phase widens scope only after the previous phase has been reviewed, which allows lessons from pilots to shape state and national integration. Phases 2 to 5 are proposed and not yet executed.

Table 24. Roadmap phases, focus and exit review

Phase	Focus	Key activities	Exit review
Phase 1 — Prototype	Demonstration of the end-to-end concept	Workflow, RBAC, GIS view, anomaly detection module, dashboards on sample data	Stakeholder demonstration and feedback
Phase 2 — Pilot	Real-world validation	Pilot with selected district / project; data migration; user training; security review	Pilot evaluation report
Phase 3 — State Integration	State-level deployment	State land record, cadastral and payment-system integration; state rule configuration	State acceptance and audit
Phase 4 — Multi-State Deployment	Replication	Onboard additional states; multilingual interface; common data standards	Cross-state review
Phase 5 — National Rollout	National operation	National dashboards, full interoperability, capacity building, support model	National governance review

Implementation status note: This document distinguishes the BhoomiSetu prototype scope from the proposed production architecture. Integrations with state land records, BhuNaksha, Bhuvan, PFMS, DILRMP, DigiLocker / eSign and LGD, as well as government-cloud hosting, are **Proposed / Integration Ready** and are not claimed to be live.

14 Impact, Benefits & Comparison

Table 25. Benefits by stakeholder

Stakeholder	Expected benefit
Central Ministries	National, comparable and current view of progress; early identification of stalled projects and states.
State Governments	Single view of all districts, faster approval handling, consistent statutory compliance tracking.
District Administration	Case pipeline visibility, automated reminders, fewer manual registers and reports.
Land Acquisition Officers	Less duplicate entry, guided stage workflow, reviewable valuation flags with reasons.
PIAs	Clear status of land availability, proposed vs. acquired land and possession for their projects.
Citizens and Landowners	Access to own parcel and case status, notices and objection channel.
Affected Families	Traceable compensation and R&R entitlement status through to closure.

Table 26. Comparison: conventional process, existing state portals and BhoomiSetu

Parameter	Conventional process	Existing state portals	BhoomiSetu
End-to-end lifecycle	Paper-based, stage by stage	Often partial; varies by state	Unified 12-stage lifecycle
GIS	Separate maps, manual overlay	Limited or separate	Integrated geo-tagging and parcel maps
Multi-stakeholder governance	File hand-offs between offices	Mostly single-department modules	Six-role RBAC with shared case record
Compensation	Manual assessment and registers	Varies; limited anomaly checks	Tracking plus AI-assisted flagging (decision support)
PFMS	Offline payment follow-up	Varies by state	PFMS-ready design (proposed integration)
R&R	Manual registers	Limited tracking	Family-level R&R tracking to closure
Timeline alerts	Manual diary tracking	Limited	Automated milestone and delay alerts
National interoperability	None	State-specific formats	Common data model; integrations proposed
Audit trails	File notes	Varies	User, role, time and IP logged; hashes
Dashboards	Periodic manual MIS	State-level, varies	National to project dashboards
Citizen transparency	Visit office / notices	Notice display; varies	Own-case status view

The comparison is qualitative and based on typical characteristics; state portals differ and the BhoomiSetu column describes design intent.

Table 27. Expected impact (qualitative; not measured results)

Impact area	Expected outcome
Faster workflow	Automated routing and reminders reduce idle time between stages.
Transparency and accountability	Role-based status visibility and audit logs make actions traceable.
Inter-agency coordination	Ministry, state, district, PIA and citizens work from one case record.
GIS visibility	Proposed, notified, acquired and possessed land seen on one map.
Compensation and R&R monitoring	Assessed vs. disbursed amounts and R&R progress tracked continuously.
Data-driven governance	Decisions supported by current indicators instead of periodic manual reports.

These are expected outcomes of the design. No measured results or performance figures are claimed; they must be established through pilot deployment.

15 Testing, Future Scope, Conclusion & References

15.1 Testing

Table 28. Test coverage plan

Area	Focus	Area	Focus
Functional testing	Screens, forms and features behave per requirements	Document testing	Upload, versioning, hash verification, viewing
Workflow testing	Stage transitions, routing, deadlines	AI/ML testing	Data checks, scoring behaviour, reason codes, threshold review
RBAC testing	Role permissions, jurisdiction scoping	Security testing	Authentication, authorisation, input handling, logging
GIS testing	Coordinate validity, overlays, status maps	Dashboard testing	Indicator accuracy and role-wise views
API testing	Request / response, errors, authorisation	Mobile responsiveness	Layout and usability on small screens
Integration testing	Component and (when available) external interfaces		

15.2 Future Scope

- Nationwide deployment
- Advanced GIS analytics
- More government integrations
- Mobile field application
- Offline field synchronization
- Advanced predictive analytics
- Multilingual expansion
- Advanced AI explainability
- More automated compliance verification
- Expanded citizen services

15.3 Conclusion

BhoomiSetu provides a unified, transparent, GIS-enabled, AI-assisted and data-driven approach to managing India's land acquisition lifecycle. By combining a 12-stage workflow, role-based governance, geo-tagged parcels, explainable decision support and multi-level dashboards, it connects land, data and decisions in real time. Its phased roadmap lets the platform grow from prototype to pilot, state integration and national rollout, while keeping final decisions with authorised government officers.

15.4 References

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